

DGR LAB2 EtherChannel v1.6 - Jorge Parra Hidalgo ITIT 13104

Antes de EtherChannel

Switch	VLAN	MAC Address	Type	Interface Física
CORE-SWITCH-JORGE1	10	00e0.b023.e017	DYNAMIC	Gig1/0/2
CORE-SWITCH-JORGE1	190	00e0.a3ae.d901	DYNAMIC	Gig1/0/20
CORE-SWITCH-JORGE1	180	00e0.a3ae.d902	DYNAMIC	Gig1/0/20
CORE-SWITCH-JORGE1	360	00e0.a3ae.d904	DYNAMIC	Gig1/0/20
ACCESS-SWITCH-JORGE1	10	90.2b8c.7d01	DYNAMIC	Fa0/24
ACCESS-SWITCH-JORGE1	190	00e0.b023.e002	DYNAMIC	Fa0/23
ACCESS-SWITCH-JORGE1	180	000a.4155.4702	DYNAMIC	Fa0/23
ACCESS-SWITCH-JORGE1	27	000a.4155.4703	DYNAMIC	Fa0/23
ACCESS-SWITCH-JORGE1	360	000a.4155.4704	DYNAMIC	Fa0/23
ACCESS-SWITCH-JORGE2	10	90.2b8c.7d02	DYNAMIC	Fa0/23
ACCESS-SWITCH-JORGE2	190	000a.4155.4701	DYNAMIC	Fa0/24
ACCESS-SWITCH-JORGE2	180	000a.4155.4702	DYNAMIC	Fa0/24
ACCESS-SWITCH-JORGE2	27	000a.4155.4703	DYNAMIC	Fa0/24
ACCESS-SWITCH-JORGE2	360	00e0.a3ae.d904	DYNAMIC	Fa0/24

Tabla 1: MAC Address Table (Antes de EtherChannel)

Switch	VLAN	MAC Address	Type	Interface Física
CORE-SWITCH-JORGE1	10	00e0.b023.e017	DYNAMIC	Gig1/0/2
CORE-SWITCH-JORGE1	190	00e0.a3ae.d901	DYNAMIC	Gig1/0/20
CORE-SWITCH-JORGE1	180	00e0.a3ae.d902	DYNAMIC	Gig1/0/20
CORE-SWITCH-JORGE1	360	00e0.a3ae.d904	DYNAMIC	Gig1/0/20
ACCESS-SWITCH-JORGE1	10	90.2b8c.7d01	DYNAMIC	Fa0/24
ACCESS-SWITCH-JORGE1	190	00e0.b023.e002	DYNAMIC	Fa0/23
ACCESS-SWITCH-JORGE1	180	000a.4155.4702	DYNAMIC	Fa0/23
ACCESS-SWITCH-JORGE1	27	000a.4155.4703	DYNAMIC	Fa0/23
ACCESS-SWITCH-JORGE1	360	000a.4155.4704	DYNAMIC	Fa0/23
ACCESS-SWITCH-JORGE2	10	90.2b8c.7d02	DYNAMIC	Fa0/23
ACCESS-SWITCH-JORGE2	190	000a.4155.4701	DYNAMIC	Fa0/24
ACCESS-SWITCH-JORGE2	180	000a.4155.4702	DYNAMIC	Fa0/24
ACCESS-SWITCH-JORGE2	27	000a.4155.4703	DYNAMIC	Fa0/24
ACCESS-SWITCH-JORGE2	360	00e0.a3ae.d904	DYNAMIC	Fa0/24

JORGE2270 000a.4155.4703 DYNAMICFa0/24**ACCESS-SWITCH-**
JORGE2360 00e0.a3ae.d904 DYNAMICFa0/24

Tabla 2: Spanning Tree VLAN 90 (Antes de EtherChannel)

Device	Interface	STP Mode / Priority	Port Status	Role	Remote / Connected Device
CORE-SWITCH-JORGE1	Gi1/0/1	RSTP / 128.1	FWD	Desg	DIST-SWITCH-JORGE1 / DIST-SWITCH-JORGE2
CORE-SWITCH-JORGE1	Gi1/0/2	RSTP / 128.2	FWD	Desg	DIST-SWITCH-JORGE1 / DIST-SWITCH-JORGE2
CORE-SWITCH-JORGE1	Gi1/0/20	RSTP / 128.20	FWD	Desg	CORE-SWITCH-JORGE2
CORE-SWITCH-JORGE1	Gi1/0/21	RSTP / 128.21	FWD	Desg	CORE-SWITCH-JORGE2
CORE-SWITCH-JORGE2	Gi1/0/20	RSTP / 128.20	FWD	Root	CORE-SWITCH-JORGE1
CORE-SWITCH-JORGE2	Gi1/0/21	RSTP / 128.21	BLK	Altn	CORE-SWITCH-JORGE1
CORE-SWITCH-JORGE2	Gi1/0/1	RSTP / 128.1	FWD	Desg	DIST-SWITCH-JORGE1 / DIST-SWITCH-JORGE2
CORE-SWITCH-JORGE2	Gi1/0/2	RSTP / 128.2	FWD	Desg	DIST-SWITCH-JORGE1 / DIST-SWITCH-JORGE2
DIST-SWITCH-JORGE1	Gi1/0/24	RSTP / 128.24	FWD	Root	CORE-SWITCH-JORGE1
DIST-SWITCH-JORGE1	Gi1/0/23	RSTP / 128.23	BLK	Altn	CORE-SWITCH-JORGE1
DIST-SWITCH-JORGE1	Gi1/0/10	RSTP / 128.10	FWD	Desg	DIST-SWITCH-JORGE2
DIST-SWITCH-JORGE1	Gi1/0/11	RSTP / 128.11	FWD	Desg	DIST-SWITCH-JORGE2
DIST-SWITCH-JORGE1	Gi1/0/1	RSTP / 128.1	FWD	Desg	ACCESS-SWITCH-JORGE2
DIST-SWITCH-JORGE1	Gi1/0/2	RSTP / 128.2	FWD	Desg	ACCESS-SWITCH-JORGE1
DIST-SWITCH-JORGE2	Gi1/0/24	RSTP / 128.24	FWD	Root	CORE-SWITCH-JORGE1 / CORE-SWITCH-JORGE2
DIST-SWITCH-JORGE2	Gi1/0/23	RSTP / 128.23	BLK	Altn	CORE-SWITCH-JORGE1 / CORE-SWITCH-JORGE2
DIST-SWITCH-JORGE2	Gi1/0/10	RSTP / 128.10	BLK	Altn	DIST-SWITCH-JORGE1
DIST-SWITCH-JORGE2	Gi1/0/11	RSTP / 128.11	BLK	Altn	DIST-SWITCH-JORGE1
DIST-SWITCH-JORGE2	Gi1/0/1	RSTP / 128.1	FWD	Desg	ACCESS-SWITCH-JORGE1
DIST-SWITCH-JORGE2	Gi1/0/2	RSTP / 128.2	FWD	Desg	ACCESS-SWITCH-JORGE2
ACCESS-SWITCH-JORGE1	Fa0/23	RSTP / 128.23	FWD	Root	DIST-SWITCH-JORGE1
ACCESS-SWITCH-JORGE1	Fa0/24	RSTP / 128.24	BLK	Altn	DIST-SWITCH-JORGE2
ACCESS-SWITCH-JORGE1	Fa0/1	RSTP / 128.1	FWD	Desg	PC1
ACCESS-SWITCH-JORGE2	Fa0/24	RSTP / 128.24	FWD	Root	DIST-SWITCH-JORGE1
ACCESS-SWITCH-JORGE2	Fa0/23	RSTP / 128.23	BLK	Altn	DIST-SWITCH-JORGE2

Root Bridge actual para VLAN 90: **CORE-SWITCH-JORGE1** (Priority: 24666 / MAC: 000A.4155.47B3).

Device	Interface	STP Mode / Priority	Port Status	Role	Remote / Connected Device
CORE-SWITCH-JORGE1	Gi1/0/1	RSTP / 128.1	FWD	Desg	DIST-SWITCH-JORGE1 / DIST-SWITCH-JORGE2
CORE-SWITCH-JORGE1	Gi1/0/2	RSTP / 128.2	FWD	Desg	DIST-SWITCH-JORGE1 / DIST-SWITCH-JORGE2
CORE-SWITCH-JORGE1	Gi1/0/20	RSTP / 128.20	FWD	Desg	CORE-SWITCH-JORGE2
CORE-SWITCH-JORGE1	Gi1/0/21	RSTP / 128.21	FWD	Desg	CORE-SWITCH-JORGE2
CORE-SWITCH-JORGE2	Gi1/0/20	RSTP / 128.20	FWD	Root	CORE-SWITCH-JORGE1
CORE-SWITCH-JORGE2	Gi1/0/21	RSTP / 128.21	BLK	Altn	CORE-SWITCH-JORGE1
CORE-SWITCH-JORGE2	Gi1/0/1	RSTP / 128.1	FWD	Desg	DIST-SWITCH-JORGE1 / DIST-SWITCH-JORGE2
CORE-SWITCH-JORGE2	Gi1/0/2	RSTP / 128.2	FWD	Desg	DIST-SWITCH-JORGE1 / DIST-SWITCH-JORGE2
DIST-SWITCH-JORGE1	Gi1/0/24	RSTP / 128.24	FWD	Root	CORE-SWITCH-JORGE1 / CORE-SWITCH-JORGE2
DIST-SWITCH-JORGE1	Gi1/0/23	RSTP / 128.23	BLK	Altn	CORE-SWITCH-JORGE1 / CORE-SWITCH-JORGE2
DIST-SWITCH-JORGE1	Gi1/0/10	RSTP / 128.10	BLK	Altn	DIST-SWITCH-JORGE1
DIST-SWITCH-JORGE1	Gi1/0/11	RSTP / 128.11	BLK	Altn	DIST-SWITCH-JORGE1
DIST-SWITCH-JORGE1	Gi1/0/1	RSTP / 128.1	FWD	Desg	ACCESS-SWITCH-JORGE1
DIST-SWITCH-JORGE1	Gi1/0/2	RSTP / 128.2	FWD	Desg	ACCESS-SWITCH-JORGE2
DIST-SWITCH-JORGE2	Gi1/0/24	RSTP / 128.24	FWD	Root	CORE-SWITCH-JORGE1 / CORE-SWITCH-JORGE2
DIST-SWITCH-JORGE2	Gi1/0/23	RSTP / 128.23	BLK	Altn	CORE-SWITCH-JORGE1 / CORE-SWITCH-JORGE2
DIST-SWITCH-JORGE2	Gi1/0/10	RSTP / 128.10	BLK	Altn	DIST-SWITCH-JORGE1
DIST-SWITCH-JORGE2	Gi1/0/11	RSTP / 128.11	BLK	Altn	DIST-SWITCH-JORGE1
DIST-SWITCH-JORGE2	Gi1/0/1	RSTP / 128.1	FWD	Desg	ACCESS-SWITCH-JORGE1
DIST-SWITCH-JORGE2	Gi1/0/2	RSTP / 128.2	FWD	Desg	ACCESS-SWITCH-JORGE2
ACCESS-SWITCH-JORGE1	Fa0/23	RSTP / 128.23	FWD	Root	DIST-SWITCH-JORGE1
ACCESS-SWITCH-JORGE1	Fa0/24	RSTP / 128.24	BLK	Altn	DIST-SWITCH-JORGE2
ACCESS-SWITCH-JORGE1	Fa0/1	RSTP / 128.1	FWD	Desg	PC1
ACCESS-SWITCH-JORGE2	Fa0/24	RSTP / 128.24	FWD	Root	DIST-SWITCH-JORGE1
ACCESS-SWITCH-JORGE2	Fa0/23	RSTP / 128.23	BLK	Altn	DIST-SWITCH-JORGE2

128.24FWDRootCORE-SWITCH-JORGE1**DIST-SWITCH-JORGE1**Gi1/0/23RSTP /
 128.23**BLK**AltnCORE-SWITCH-JORGE1**DIST-SWITCH-JORGE1**Gi1/0/10RSTP /
 128.10FWDDesgDIST-SWITCH-JORGE2**DIST-SWITCH-JORGE1**Gi1/0/11RSTP /
 128.11FWDDesgDIST-SWITCH-JORGE2**DIST-SWITCH-JORGE1**Gi1/0/1RSTP /
 128.1FWDDesgACCESS-SWITCH-JORGE2**DIST-SWITCH-JORGE1**Gi1/0/2RSTP /
 128.2FWDDesgACCESS-SWITCH-JORGE1**DIST-SWITCH-JORGE2**Gi1/0/24RSTP /
 128.24FWDRootCORE-SWITCH-JORGE1 / CORE-SWITCH-JORGE2**DIST-SWITCH-**
JORGE2Gi1/0/23RSTP / 128.23**BLK**AltnCORE-SWITCH-JORGE1 / CORE-SWITCH-
 JORGE2**DIST-SWITCH-JORGE2**Gi1/0/10RSTP / 128.10**BLK**AltnDIST-SWITCH-
 JORGE1**DIST-SWITCH-JORGE2**Gi1/0/11RSTP / 128.11**BLK**AltnDIST-SWITCH-
 JORGE1**DIST-SWITCH-JORGE2**Gi1/0/1RSTP / 128.1FWDDesgACCESS-SWITCH-
 JORGE1**DIST-SWITCH-JORGE2**Gi1/0/2RSTP / 128.2FWDDesgACCESS-SWITCH-
 JORGE2**ACCESS-SWITCH-JORGE1**Fa0/23RSTP / 128.23FWDRootDIST-SWITCH-
 JORGE1**ACCESS-SWITCH-JORGE1**Fa0/24RSTP / 128.24**BLK**AltnDIST-SWITCH-
 JORGE2**ACCESS-SWITCH-JORGE1**Fa0/1RSTP / 128.1FWDDesgPC1**ACCESS-**
SWITCH-JORGE2Fa0/24RSTP / 128.24FWDRootDIST-SWITCH-JORGE1**ACCESS-**
SWITCH-JORGE2Fa0/23RSTP / 128.23**BLK**

Configuración de EtherChannel

```

ACCESS-SWITCH-JORGE1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ACCESS-SWITCH-JORGE1(config)#interface range FastEthernet 0/23 - 24
ACCESS-SWITCH-JORGE1(config-if-range)# channel-group 1 mode active
%EC-5-L3DONTBNDL2: Fa0/23 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/24 suspended: LACP currently not enabled on the remote port.
ACCESS-SWITCH-JORGE1(config-if-range)#exit
ACCESS-SWITCH-JORGE1(config)#interface Port-channel 1
ACCESS-SWITCH-JORGE1(config-if)# switchport mode trunk
ACCESS-SWITCH-JORGE1(config-if)# switchport trunk allowed vlan 90,180,270,360
ACCESS-SWITCH-JORGE1(config-if)#end
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

```

```

ACCESS-SWITCH-JORGE2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ACCESS-SWITCH-JORGE2(config)#interface range FastEthernet 0/23 - 24
ACCESS-SWITCH-JORGE2(config-if-range)# channel-group 2 mode active
%EC-5-L3DONTBNDL2: Fa0/23 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/24 suspended: LACP currently not enabled on the remote port.
ACCESS-SWITCH-JORGE2(config-if-range)#exit
ACCESS-SWITCH-JORGE2(config)#interface Port-channel 2
ACCESS-SWITCH-JORGE2(config-if)# switchport mode trunk
ACCESS-SWITCH-JORGE2(config-if)# switchport trunk allowed vlan 90,180,270,360
ACCESS-SWITCH-JORGE2(config-if)#end
Creating a port-channel interface Port-channel 2

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

```

```

DIST-SWITCH-JORGE1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
DIST-SWITCH-JORGE1(config)#
DIST-SWITCH-JORGE1(config)#! 1. Alineacin para Port-channel 1 (hacia ACCESS1)
DIST-SWITCH-JORGE1(config)#interface GigabitEthernet 1/0/2
DIST-SWITCH-JORGE1(config-if)# switchport mode trunk
DIST-SWITCH-JORGE1(config-if)# switchport trunk allowed vlan 90,180,270,360
DIST-SWITCH-JORGE1(config-if)# channel-group 1 mode active
DIST-SWITCH-JORGE1(config-if)#exit
DIST-SWITCH-JORGE1(config)#
DIST-SWITCH-JORGE1(config)#! 2. Alineacin para Port-channel 2 (hacia ACCESS2)
DIST-SWITCH-JORGE1(config)#interface GigabitEthernet 1/0/1
DIST-SWITCH-JORGE1(config-if)# switchport mode trunk
DIST-SWITCH-JORGE1(config-if)# switchport trunk allowed vlan 90,180,270,360
DIST-SWITCH-JORGE1(config-if)# channel-group 2 mode active
DIST-SWITCH-JORGE1(config-if)#exit
DIST-SWITCH-JORGE1(config)#
DIST-SWITCH-JORGE1(config)#! 3. Alineacin para Port-channel 3 (hacia DIST2)
DIST-SWITCH-JORGE1(config)#interface range GigabitEthernet 1/0/10 - 11
DIST-SWITCH-JORGE1(config-if-range)# switchport mode trunk
DIST-SWITCH-JORGE1(config-if-range)# switchport trunk allowed vlan 90,180,270,360
DIST-SWITCH-JORGE1(config-if-range)# channel-group 3 mode active
DIST-SWITCH-JORGE1(config-if-range)#exit
DIST-SWITCH-JORGE1(config)#
DIST-SWITCH-JORGE1(config)#! 4. Alineacin para Port-channel 4 (hacia CORE1)
DIST-SWITCH-JORGE1(config)#interface range GigabitEthernet 1/0/23 - 24
DIST-SWITCH-JORGE1(config-if-range)# switchport mode trunk
DIST-SWITCH-JORGE1(config-if-range)# switchport trunk allowed vlan 90,180,270,360
DIST-SWITCH-JORGE1(config-if-range)# channel-group 4 mode active
DIST-SWITCH-JORGE1(config-if-range)#end
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to
down

```



```

DIST-SWITCH-JORGE2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
DIST-SWITCH-JORGE2(config)#
DIST-SWITCH-JORGE2(config)#! Port-channel 1 (hacia ACCESS1)
DIST-SWITCH-JORGE2(config)#interface GigabitEthernet 1/0/1
DIST-SWITCH-JORGE2(config-if)# switchport mode trunk
DIST-SWITCH-JORGE2(config-if)# switchport trunk allowed vlan 90,180,270,360
DIST-SWITCH-JORGE2(config-if)# channel-group 1 mode active
%EC-5-L3DONTBNDL2: Gig1/0/1 suspended: LACP currently not enabled on the remote port.
DIST-SWITCH-JORGE2(config-if)#exit
DIST-SWITCH-JORGE2(config)#interface Port-channel 1
DIST-SWITCH-JORGE2(config-if)# switchport mode trunk
DIST-SWITCH-JORGE2(config-if)# switchport trunk allowed vlan 90,180,270,360
DIST-SWITCH-JORGE2(config-if)#exit
DIST-SWITCH-JORGE2(config)#
DIST-SWITCH-JORGE2(config)#! Port-channel 2 (hacia ACCESS2)
DIST-SWITCH-JORGE2(config)#interface GigabitEthernet 1/0/2
DIST-SWITCH-JORGE2(config-if)# switchport mode trunk
DIST-SWITCH-JORGE2(config-if)# switchport trunk allowed vlan 90,180,270,360
DIST-SWITCH-JORGE2(config-if)# channel-group 2 mode active
%EC-5-L3DONTBNDL2: Gig1/0/2 suspended: LACP currently not enabled on the remote port.
DIST-SWITCH-JORGE2(config-if)#exit
DIST-SWITCH-JORGE2(config)#interface Port-channel 2
DIST-SWITCH-JORGE2(config-if)# switchport mode trunk
DIST-SWITCH-JORGE2(config-if)# switchport trunk allowed vlan 90,180,270,360
DIST-SWITCH-JORGE2(config-if)#exit
DIST-SWITCH-JORGE2(config)#
DIST-SWITCH-JORGE2(config)#! Port-channel 3 (hacia DIST1)
DIST-SWITCH-JORGE2(config)#interface range GigabitEthernet 1/0/10 - 11
DIST-SWITCH-JORGE2(config-if-range)# switchport mode trunk
DIST-SWITCH-JORGE2(config-if-range)# switchport trunk allowed vlan 90,180,270,360
DIST-SWITCH-JORGE2(config-if-range)# channel-group 3 mode active
%EC-5-L3DONTBNDL2: Gig1/0/10 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Gig1/0/11 suspended: LACP currently not enabled on the remote port.
DIST-SWITCH-JORGE2(config-if-range)#exit
DIST-SWITCH-JORGE2(config)#interface Port-channel 3

```

```

CORE-SWITCH-JORGE1#
CORE-SWITCH-JORGE1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
CORE-SWITCH-JORGE1(config)#
CORE-SWITCH-JORGE1(config)#! 1. Forzar alineacin en Port-channel 4
CORE-SWITCH-JORGE1(config)#interface range GigabitEthernet 1/0/1 - 2
CORE-SWITCH-JORGE1(config-if-range)# switchport mode trunk
CORE-SWITCH-JORGE1(config-if-range)# switchport trunk allowed vlan 90,180,270,360
CORE-SWITCH-JORGE1(config-if-range)# channel-group 4 mode active
CORE-SWITCH-JORGE1(config-if-range)#exit
CORE-SWITCH-JORGE1(config)#
CORE-SWITCH-JORGE1(config)#! 2. Forzar alineacin en Port-channel 6
CORE-SWITCH-JORGE1(config)#interface range GigabitEthernet 1/0/20 - 21
CORE-SWITCH-JORGE1(config-if-range)# switchport mode trunk
CORE-SWITCH-JORGE1(config-if-range)# switchport trunk allowed vlan 90,180,270,360
CORE-SWITCH-JORGE1(config-if-range)# channel-group 6 mode active
CORE-SWITCH-JORGE1(config-if-range)#end
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan90, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan180, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan270, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan360, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to
down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to up

```

```

CORE-SWITCH-JORGE2#
CORE-SWITCH-JORGE2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
CORE-SWITCH-JORGE2(config)#
CORE-SWITCH-JORGE2(config)#! 1. Configurar Port-channel 5 (hacia DIST2)
CORE-SWITCH-JORGE2(config)#interface range GigabitEthernet 1/0/1 - 2
CORE-SWITCH-JORGE2(config-if-range)# switchport mode trunk
CORE-SWITCH-JORGE2(config-if-range)# switchport trunk allowed vlan 90,180,270,360
CORE-SWITCH-JORGE2(config-if-range)# channel-group 5 mode active
%EC-5-L3DONTBNDL2: Gig1/0/1 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Gig1/0/2 suspended: LACP currently not enabled on the remote port.
CORE-SWITCH-JORGE2(config-if-range)#exit
CORE-SWITCH-JORGE2(config)#interface Port-channel 5
CORE-SWITCH-JORGE2(config-if)# switchport mode trunk
CORE-SWITCH-JORGE2(config-if)# switchport trunk allowed vlan 90,180,270,360
CORE-SWITCH-JORGE2(config-if)#exit
CORE-SWITCH-JORGE2(config)#
CORE-SWITCH-JORGE2(config)#! 2. Configurar Port-channel 6 (hacia CORE1)
CORE-SWITCH-JORGE2(config)#interface range GigabitEthernet 1/0/20 - 21
CORE-SWITCH-JORGE2(config-if-range)# switchport mode trunk
CORE-SWITCH-JORGE2(config-if-range)# switchport trunk allowed vlan 90,180,270,360
CORE-SWITCH-JORGE2(config-if-range)# channel-group 6 mode active
%EC-5-L3DONTBNDL2: Gig1/0/20 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Gig1/0/21 suspended: LACP currently not enabled on the remote port.
CORE-SWITCH-JORGE2(config-if-range)#exit
CORE-SWITCH-JORGE2(config)#interface Port-channel 6
CORE-SWITCH-JORGE2(config-if)# switchport mode trunk
CORE-SWITCH-JORGE2(config-if)# switchport trunk allowed vlan 90,180,270,360
CORE-SWITCH-JORGE2(config-if)#end
Creating a port-channel interface Port-channel 5

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to
down

%EC-5-CANNOT_BUNDLE2: Gig1/0/1 is not compatible with Po5 and will be suspended (vlan
mask is different)

```

show etherchannel summary

```

CORE-SWITCH-JORGE1#show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 2
Number of aggregators:          2

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
4      Po4(SU)        LACP       Gig1/0/1(P) Gig1/0/2(P)
6      Po6(SU)        LACP       Gig1/0/20(P) Gig1/0/21(P)
CORE-SWITCH-JORGE1#

```



```

CORE-SWITCH-JORGE2#show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 2
Number of aggregators:          2

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
5      Po5(SU)        LACP        Gig1/0/1(P) Gig1/0/2(P)
6      Po6(SU)        LACP        Gig1/0/20(P) Gig1/0/21(P)
CORE-SWITCH-JORGE2#
CORE-SWITCH-JORGE2#
CORE-SWITCH-JORGE2#

```

show mac address-table

```

CORE-SWITCH-JORGE1#show mac address-table
Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
360     00e0.a3ae.d904   DYNAMIC     Po6
CORE-SWITCH-JORGE1#

CORE-SWITCH-JORGE2#show mac address-table
Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
90      0005.5ec1.4214   DYNAMIC     Gig1/0/20
90      0005.5ec1.4215   DYNAMIC     Gig1/0/21
90      0090.2b8c.7d18   DYNAMIC     Gig1/0/1
CORE-SWITCH-JORGE2#

```

```

DIST-SWITCH-JORGE1#show mac address-table
Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
90      0001.63d3.d917   DYNAMIC Gig1/0/2
90      0005.5ec1.4201   DYNAMIC Gig1/0/24
90      0005.5ec1.4202   DYNAMIC Gig1/0/23
90      000a.4155.4701   DYNAMIC Po4
90      0090.2152.e818   DYNAMIC Gig1/0/1
90      0090.2b5b.d51b   DYNAMIC Po4
90      00e0.a3ae.d901   DYNAMIC Po4
180     000a.4155.4702   DYNAMIC Po4
180     0090.2b5b.d51b   DYNAMIC Po4
180     00e0.a3ae.d902   DYNAMIC Po4
270     000a.4155.4703   DYNAMIC Po4
270     0090.2b5b.d51b   DYNAMIC Po4
270     00e0.a3ae.d903   DYNAMIC Po4
DIST-SWITCH-JORGE1#
DIST-SWITCH-JORGE1#

```

```

DIST-SWITCH-JORGE2#
DIST-SWITCH-JORGE2#show mac address-table
Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
90      0001.63d3.d918   DYNAMIC Gig1/0/1
90      000a.4155.4701   DYNAMIC Po5
90      000a.415e.48a9   DYNAMIC Po5
90      000a.f3a1.e601   DYNAMIC Gig1/0/24
90      000a.f3a1.e602   DYNAMIC Gig1/0/23
90      0090.2152.e817   DYNAMIC Gig1/0/2
90      00e0.a3ae.d901   DYNAMIC Po5
180     000a.4155.4702   DYNAMIC Po5
270     000a.4155.4703   DYNAMIC Po5
270     000a.415e.48a9   DYNAMIC Po5
360     000a.4155.4704   DYNAMIC Po5
360     000a.415e.48a9   DYNAMIC Po5
360     00e0.a3ae.d904   DYNAMIC Po5
DIST-SWITCH-JORGE2#

```

show spanning-tree vlan 90

```

DIST-SWITCH-JORGE1#show spanning-tree vlan 90
VLAN0090
  Spanning tree enabled protocol rstp
  Root ID    Priority    24666
             Address     000A.4155.47B3
             Cost        3
             Port        32 (Port-channel4)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32858  (priority 32768 sys-id-ext 90)
             Address     00D0.BA15.B675
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Po2                      Desg FWD 19        128.30 P2p
Po3                      Desg FWD 3        128.31 P2p
Po4                      Root FWD 3        128.32 P2p
Po1                      Desg FWD 19       128.29 P2p

DIST-SWITCH-JORGE1#

```

```

ACCESS-SWITCH-JORGE1#VLAN0090
Translating "VLAN0090"...domain server (255.255.255.255) % Name lookup aborted
ACCESS-SWITCH-JORGE1#show spanning-tree vlan 90
VLAN0090
  Spanning tree enabled protocol rstp
  Root ID    Priority    24666
             Address     000A.4155.47B3
             Cost        22
             Port        27 (Port-channel1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32858  (priority 32768 sys-id-ext 90)
             Address     0001.9624.54BA
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/1                   Desg FWD 19        128.1  P2p
Po2                     Altn BLK 19        128.28 P2p
Po1                     Root FWD 19        128.27 P2p

```

Elegir la ruta y preparar el ping continuo pc1 a 8.8.8.8

```

C:\>ping 8.8.8.8 -n 1000

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253

```

Interrumpir el tráfico apagando el Port-Channel completo

```
CORE-SWITCH-JORGE1>
CORE-SWITCH-JORGE1>
CORE-SWITCH-JORGE1>ena
CORE-SWITCH-JORGE1>enable
CORE-SWITCH-JORGE1#en
CORE-SWITCH-JORGE1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
CORE-SWITCH-JORGE1(config)#interface Port-channel 5
CORE-SWITCH-JORGE1(config-if)#shutdown

%LINK-5-CHANGED: Interface Port-channel5, changed state to administratively down
CORE-SWITCH-JORGE1(config-if)#
```

Restaurar el canal y probar la resiliencia (Falla de un solo enlace físico)

```
ACCESS-SWITCH-JORGE1#
ACCESS-SWITCH-JORGE1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ACCESS-SWITCH-JORGE1(config)#interface Port-channel 2
ACCESS-SWITCH-JORGE1(config-if)#shutdown

ACCESS-SWITCH-JORGE1(config-if)#end
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

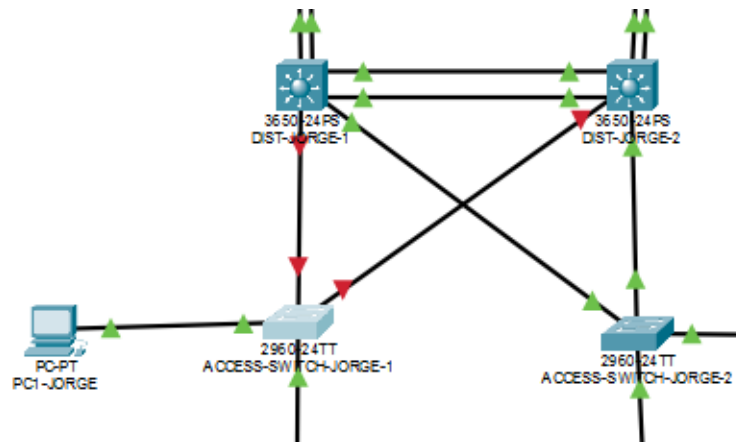
%LINK-5-CHANGED: Interface Port-channel2, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to down

ACCESS-SWITCH-JORGE1#
%SYS-5-CONFIG_I: Configured from console by console

ACCESS-SWITCH-JORGE1#
ACCESS-SWITCH-JORGE1#
ACCESS-SWITCH-JORGE1#
ACCESS-SWITCH-JORGE1#
ACCESS-SWITCH-JORGE1#
ACCESS-SWITCH-JORGE1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ACCESS-SWITCH-JORGE1(config)#interface range FastEthernet 0/1 - 24
ACCESS-SWITCH-JORGE1(config-if-range)#shutdown

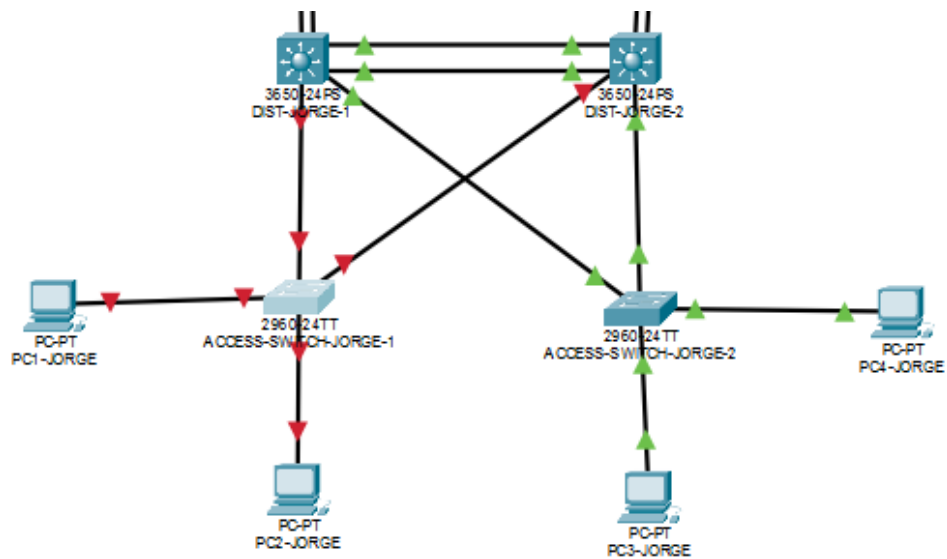
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to administratively down
```



```

Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Request timed out.
Request timed out.
Request timed out.
Request timed out.

```



Verificar el estado del Port-Channel tras la falla parcial


```

ACCESS-SWITCH-JORGE1#
ACCESS-SWITCH-JORGE1#
ACCESS-SWITCH-JORGE1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
ACCESS-SWITCH-JORGE1(config)#interface range FastEthernet 0/1 - 2
ACCESS-SWITCH-JORGE1(config-if-range)#no shutdown
ACCESS-SWITCH-JORGE1(config-if-range)#interface range FastEthernet 0/23 - 24
ACCESS-SWITCH-JORGE1(config-if-range)#no shutdown
ACCESS-SWITCH-JORGE1(config-if-range)#end
ACCESS-SWITCH-JORGE1#
%SYS-5-CONFIG_I: Configured from console by console

ACCESS-SWITCH-JORGE1#

```

```

Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Reply from 8.8.8.8: bytes=32 time=1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253
Reply from 8.8.8.8: bytes=32 time=11ms TTL=253
Reply from 8.8.8.8: bytes=32 time<1ms TTL=253

```

Resumen/Conclusion

Este laboratorio permitió validar de forma integral el diseño, la operación y la resiliencia de una infraestructura de red de alta disponibilidad orientada a entornos corporativos. A lo largo de las pruebas, se conjugaron elementos clave de conmutación avanzada, agregación de enlaces y convergencia de topologías para garantizar la continuidad operativa de los servicios de los usuarios finales.

Los puntos centrales que definieron el desarrollo y el éxito de la práctica abarcan:

- **Implementación de Agregación de Enlaces (EtherChannel / LACP):** Se configuraron canales lógicos (Po1 a Po5) interconectando los switches de acceso, distribución y core (ACCESS-SWITCH-JORGE1 , DIST-SWITCH-JORGE1/2 , CORE-SWITCH-JORGE1). Esto permitió no solo incrementar el ancho de banda disponible

entre las capas de la red, sino también unificar múltiples enlaces físicos en una sola interfaz lógica resistente a fallos individuales.

- **Control de Bucles y Gestión de Topologías con RSTP:** Mediante el uso de Rapid Spanning Tree Protocol, la red mantuvo una estructura completamente libre de bucles de capa 2. Se identificaron y validaron de manera empírica los roles de puertos (`Altn BLK` , `Root` , etc.) y las rutas de respaldo activas que garantizan una arquitectura tolerante a fallos.
- **Pruebas Empíricas de Resiliencia y Fallos:** Se ejecutaron escenarios de aislamiento y corte total de tráfico (mediante el apagado controlado de interfaces físicas y Port-channels) para demostrar la dependencia de los flujos de red sobre las rutas principales. Asimismo, se comprobó la capacidad de autorrecuperación de la topología al desviar el tráfico de forma transparente a través de los enlaces redundantes.
- **Gestión de Tiempos de Convergencia:** Se experimentaron de primera mano los efectos transitorios de los estados de convergencia del Spanning Tree (tránsito por estados de escucha y aprendizaje hacia el reenvío definitivo), comprendiendo la importancia de una correcta sintonización de los temporizadores y la administración precisa de puertos para evitar cortes prolongados de servicio.

En definitiva, la práctica demostró con éxito cómo una arquitectura redundante bien dimensionada e implementada es capaz de sobreponerse a la caída de enlaces críticos, manteniendo la estabilidad operativa y asegurando que los dispositivos de la red recuperen la conectividad de manera automática y eficiente.